re-checked the rows using value-level entity comparison, we were able see the types of loss separately. In 9.7 percent of rows, the output keeps digits, but not the annotated numeral. In 3.7 percent of rows, the output is corrupted. In 3.3 percent of rows the, output contains no digits all. 0.4 percent of rows have the same quantity, just rescaled. In 30 rows or 0.4 percent, the quantity survived, but the recovery missed it, which limits the error of the recovery procedure itself.

Table 5.3: Survival of the annotated numeral in task 1 machine-translated rows

Outcome	Rows	Share
Span recovered	56,624	82.8%
Digits present, annotated value gone	6,618	9.7%
Output corrupted (placeholder or escape)	2,560	3.7%
No digits in the output	2,234	3.3%
Same quantity, rescaled	292	0.4%
Quantity preserved, span missed	30	0.04%

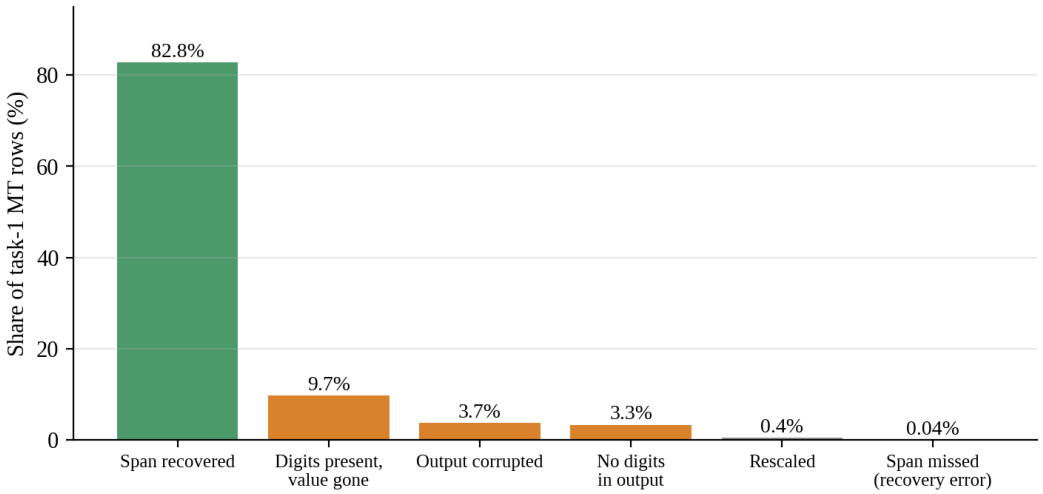


Figure 5.1: Outcome of the annotated numeral in task 1 machine translation

About one out of every six financial numerals does not remain unchanged during Indic-to-Indic translation. A stricter count includes every currency symbol and percentage that appears in a sentence, not just the one marked. Rows that are corrupted, which represent 6.2 percent of task 1, are omitted from this calculation. Still, no direction in task 1 reaches the 95 percent threshold: the rates vary from 49.8 percent for the Bengali-to-Malayalam direction, to 78.7 percent for the Telugu-to-Hindi direction (see Figure 5.2). Directions in task 2 range from 89.5

percent to 95.7 percent, while directions in task 3 range from 77.4 percent to 97.0-percent. In each set, the weakest direction is Hindi-to-Malayalam.

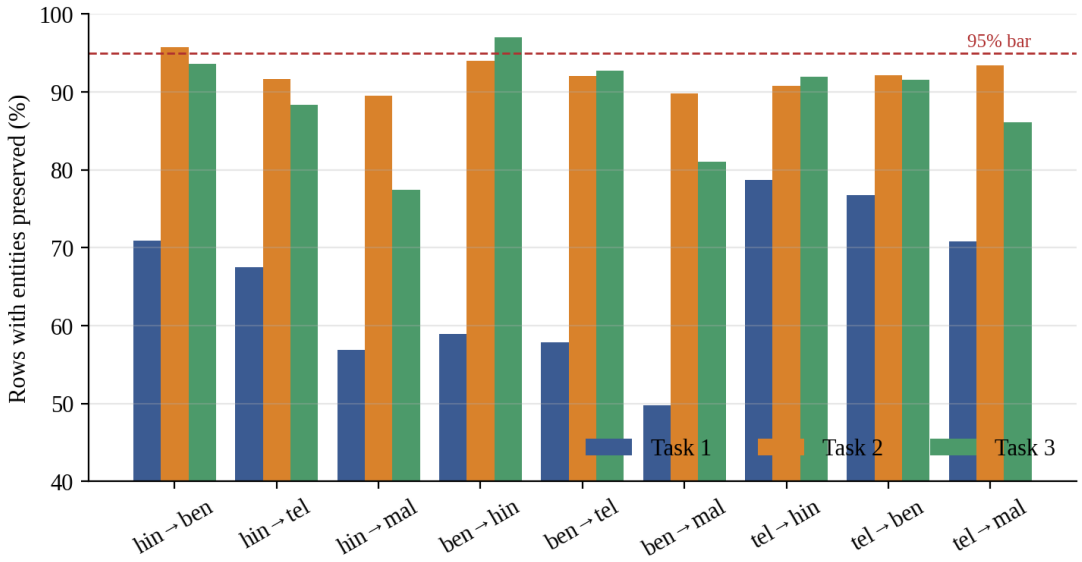


Figure 5.2: Entity preservation by direction and task, with the 95 percent bar

5.4.3 SEMANTIC SIMILARITY BY DIRECTION

In the LaBSE similarity analysis, task 1 is split by source language (see Table 5.4, Figure 5.3). Hindi-source directions show a similarity ranging from 0.906 to 0.937, and they dip below τ on between 5 and 11 percent of the rows. Bengali-source directions show a similarity ranging from 0.844 to 0.854, and they dip below τ on between 29 and 34 percent of the rows. This pattern shows up in task 1, the only task that has blocks with different sentences so it reflects the source texts as much as the translation directions.

Table 5.4: Median LaBSE similarity and share below τ by direction

Direction	T1 median	T1 below τ	T2 median	T2 below τ	T3 median	T3 below τ
hin → ben	0.935	6.5%	0.950	0.3%	0.908	0.4%
hin → tel	0.937	4.7%	0.945	0.2%	0.903	0.8%
hin → mal	0.906	10.9%	0.908	4.4%	0.859	4.5%
ben → hin	0.854	29.4%	0.931	0.8%	0.898	0.4%
ben → tel	0.849	31.5%	0.923	2.4%	0.889	1.1%
ben → mal	0.844	34.1%	0.902	5.4%	0.870	2.1%
tel → hin	0.916	15.5%	0.943	0.4%	0.891	1.1%
tel → ben	0.909	19.8%	0.939	0.7%	0.881	1.9%
tel → mal	0.906	18.8%	0.924	1.6%	0.861	3.8%

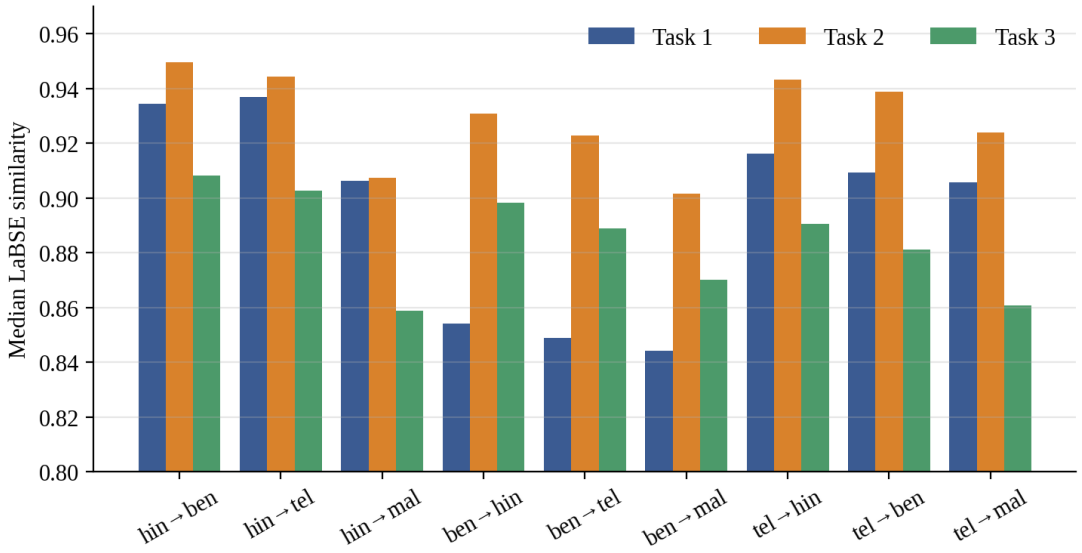


Figure 5.3: Median LaBSE similarity by translation direction and task

5.4.4 CALIBRATING THE THRESHOLD

Human translations in task 2 and task 3 allow us to check the threshold. In task 2, the median score for pairs is 0.921 and the 5th percentile is 0.846. With τ equal to 0.82, every human translation in task 2 passes the threshold. In task 3, the median score is 0.870 and the 5th percentile is 0.729. Using τ equal to 0.82 would cause 23.4 percent of translations in task 3 to fail the threshold. The reason for the difference is the length of the text. Task 2 has a length of 148 characters while task 3 has a median length of 86 characters. LaBSE cosine similarity decreases when the context becomes shorter. Therefore, τ is set separately for each task as shown in Table 5.5. When τ is calibrated to 0.73, the number of drift-flagged rows in task 3 drops from 740 to 85.

Table 5.5: Calibrated similarity threshold by task

Task	τ	Basis
Task 1	0.82	Inherited, not calibrated, because no human reference exists in any direction
Task 2	0.82	Confirmed: rejects 0.0% of human pairs and no direction exceeds the 20% trigger
Task 3	0.73	Revised: the value that accepts 95% of human pairs

A second result qualifies every similarity figure in this report. Machine translation scores higher than translation of the same items in both calibratable tasks and in every language pair. On task 2 for example Hindi to Bengali machine translation scores 0.950 against 0.918 for the human Bengali and Hindi pair;, on task 3 the figures are

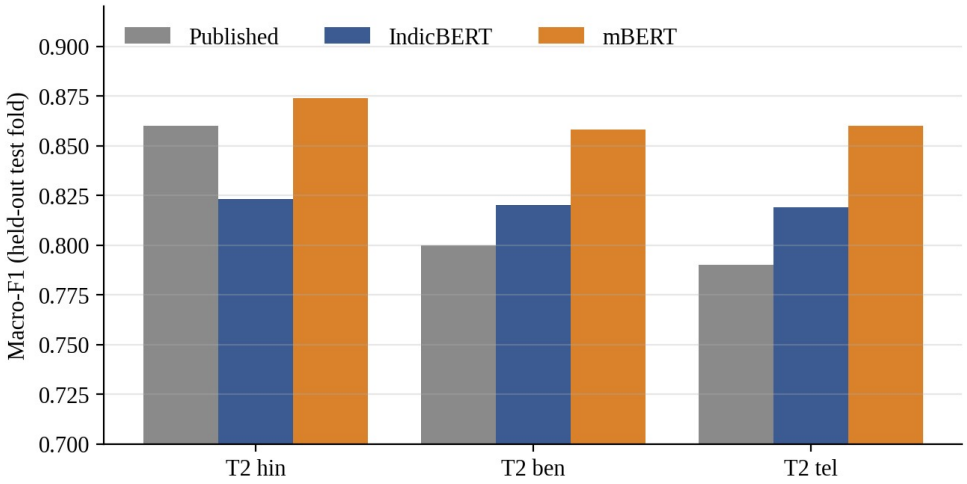


Figure 5.4: In-language macro-F1 for task 2 against the published figures

Three of the nine task 3 IndicBERT runs did not fit their training data: IndicBERT with Bengali seed 0 and IndicBERT with Telugu seeds 0 and 2 reached a peak below 0.5 for training macro-F1. On a task that has reached its data ceiling, the test scores of those runs look like those of a healthy run. This is why convergence is checked on the training fold. mBERT had no failures under the same settings. These runs still appear in every all-seeds figure. Over converged seeds, task 3 Bengali and task 3 Telugu score 0.159 and 0.150.

5.6 ZERO-SHOT TRANSFER

Two properties of the measurement matter when reading the transfer results. First, the in-language score that the gap subtracts from is computed on the evaluation items in the source language. 80 Percent of the items on task 2 and 70 percent on task 3 are in the source model's training fold. This represents a training-text ceiling. For task 2 in Hindi with IndicBERT, the score reaches 0.954, compared to 0.823 on the held-out fold. Second, task 3 is at its data ceiling with 532 headlines spread across ten classes.

Figures 5.5 and 5.6 show the gap for every source and every target. Appendix 4 gives the table with seed spread. Transfer costs a large share of in-language performance in every cell. On task 2 the gaps range from 0.26 to 0.45 with IndicBERT and from 0.11 to 0.24 with mBERT. Out of Hindi, IndicBERT loses 0.39 to 0.45 on task 2 and 0.50 to 0.51 on task 3. On task 3 the gaps stay between 0.46 and 0.56 no matter the source, target, or encoder.

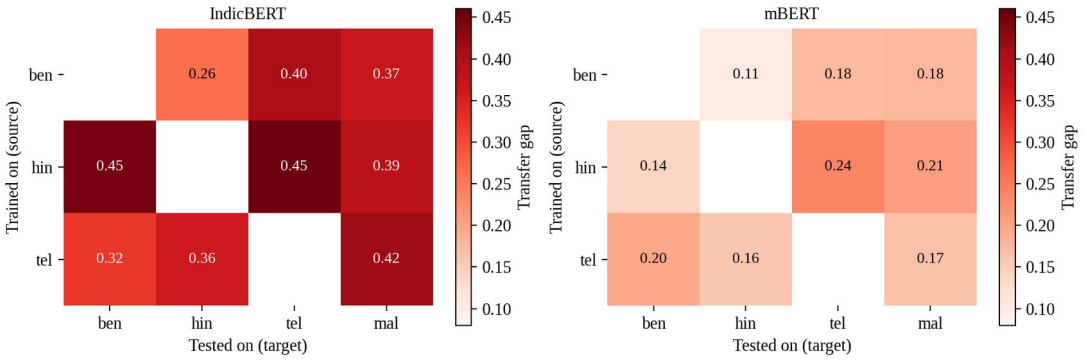


Figure 5.5: Task 2 transfer gap, source on the vertical axis and target on the horizontal axis

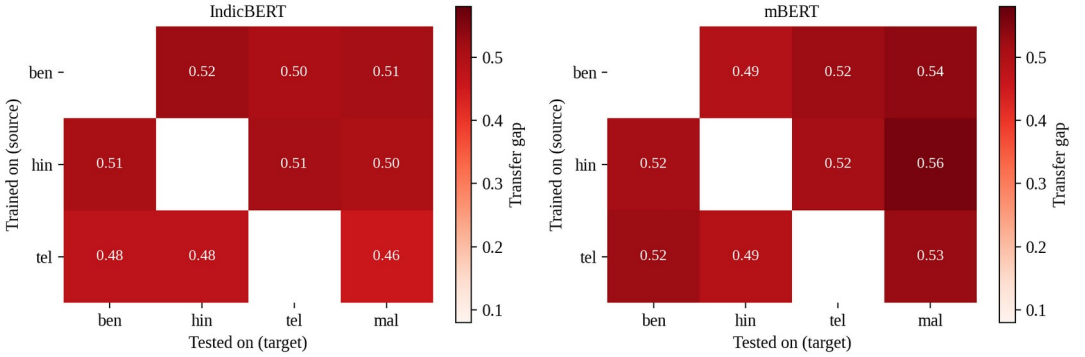


Figure 5.6: Task 3 transfer gap, source on the vertical axis and target on the horizontal axis

5.6.1 IS THE LOSS AN ARTEFACT OF THE TRANSLATIONS?

When we evaluate using machine-translated target text from the sources block instead of native text, the gaps shift by only a few points at most. For the Hindi source and IndicBERT on task two the gaps are between 0.38 and 0.47 on machine-translated targets and between 0.39 and 0.45 on targets. Thus the loss is not an artefact of this project's translations. If translation damage had been the driver, evaluation on genuine human-written Bengali would have scored noticeably better than on machine-translated Bengali but it did not.

5.6.2 TRANSLATIONESE CONDITIONS

The translationese conditions keep the language the same and only change where the text comes from. When using IndicBERT on task 2, a Hindi, Bengali or Telugu model scores between 5.7 and 7.8 points lower on machine-translated text in its language compared to native text. On task 3 the drop is even bigger ranging from 8.8 to 26.6 points. The native arm refers to the model's split, which is mostly made up of its

5.8 SOURCE LANGUAGE AND DIRECTION

Training on Hindi gives the largest mean gap on task 2 with both encoders, and Bengali the smallest (Table 5.8, Figure 5.7). The three source models are equally good in their own language, within a quarter of a point of each other, so the ranking concerns transfer and not a stronger model.

Table 5.8: Mean transfer gap by source language and target family, task 2

Group	IndicBERT	mBERT
Source: Bengali	0.345	0.153
Source: Telugu	0.367	0.174
Source: Hindi	0.429	0.196
Target family: Indo-Aryan	0.349	0.151
Target family: Dravidian	0.406	0.194

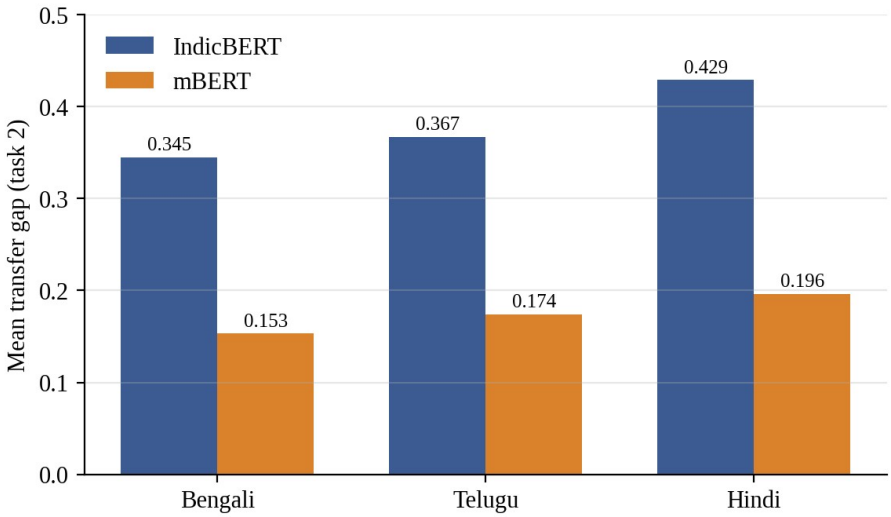


Figure 5.7: Mean transfer gap by source language on task 2

The cost is mostly in the target language. When translating into a language the mean gap is 0.406 with IndicBERT and 0.194 with mBERT. This compares to 0.349 and 0.151 when translating into a Aryan language. The corpus quality results in Section 5.4 showed the trend in translation quality measured using a different method. This agreement is the result in the study. Transfer is also not equal in both directions. Hindi to Bengali loses more than Bengali to Hindi on both encoders beyond what might be explained by seed noise: 0.448 versus 0.263 with IndicBERT and 0.142 versus 0.105, with mBERT.

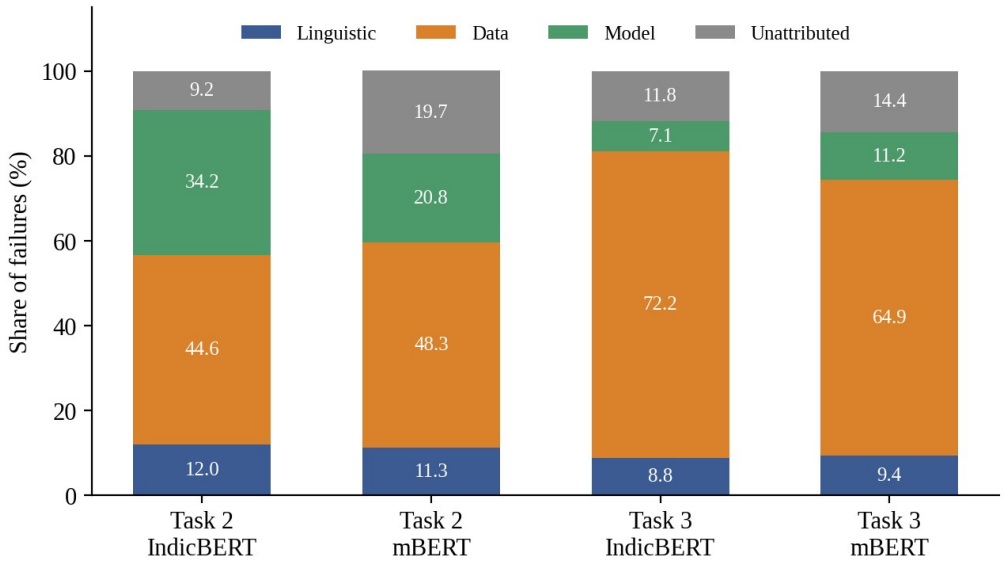


Figure 5.8: Composition of transfer failures by task and encoder

Two of the modules failed to validate. Items that contain a masked term do not fail more often than items that keep the term intact: masked items fail more often in only 20 out of 44 cells. Saliency divergence does not separate failures from handled pairs; the area under the curve is between 0.48 and 0.52. Tokenizer fragmentation never fires. At most, 0.5 percent of failures involve tokenizer fragmentation because a ratio of 2.0 is rarely reached. Labels from these modules are reported. They should not be read as established causes, however.

Of 228 group comparisons, 22 meet the finding rule, and 11 of those could be produced by a known measurement limit on its own. The remaining 11 fall into three groups.

Direction (task 2). Failures out of Telugu carry more model-behaviour labels than failures into it, on both encoders (40.2 percent against 34.4 percent with IndicBERT, 24.9 percent against 14.4 percent with mBERT), mainly through seed instability.

Source. On task 2, when the target languages are Bengali and Malayalam, failures coming from Telugu sources have more model-behaviour labels compared to those from Hindi sources. The numbers show 46.8 percent for Telugu-source failures versus 18.4 percent for Hindi-source ones. Similarly, for the comparison, it's 22.0 percent versus 14.7 percent. On task 3, Hindi-source failures have more terminology_gap labels than Telugu-source failures. This